

Syntax			
- Terms refer to objects in a domain - Predicates are used to make truth statements about objects			
Functions			
$\mathcal{F}$ is the set of all functions.			
- A function f of arity n map n terms to one term - Constants are functions of arity 0			
Terms			
- functions $f \in \mathcal{F}$ refer to objects implicitly through other terms - constants $c \in \mathcal{F}$ refer to individual objects - variables $v \in \mathcal{V}$ are placeholders for objects			
The set of terms $\mathcal{T}$ is defined inductively:			
$\forall v \in \mathcal{V} : v \in \mathcal{T}$ $\forall c \in \mathcal{F} : c \in \mathcal{T}$ $\forall f$ (of arity n) $\in \mathcal{F} : \forall t_1, \dots, t_n \in \mathcal{T} : f(t_1, \dots, t_n) \in \mathcal{T}$			
Predicates			
$\mathcal{P}$ refers to the set of predicate symbols			
- A predicate symbol $P \in \mathcal{P}$ of arity n maps n terms to an atom. - A predicate symbol $P \in \mathcal{P}$ of arity 0 acts like a propositional variable.			
Formulas			
The set of formulas is inductively defined as:			
- If φ is a formula, then $\neg\varphi$ is a formula - If φ, ψ are formulas, then $\varphi \odot \psi$ is a formula where $\odot \in \{\vee, \wedge, \rightarrow\}$ - If φ is a formula, then $\forall v \in \mathcal{V} : \mathcal{Q}x : \psi$ is a formula where $\mathcal{Q} \in \{\exists, \forall\}$			
Ground Terms/Formulas			
A term/formula is ground if it contains no variable.			
Subformulas			
- If φ is a atomic, then $subformulas(\varphi) = \{\varphi\}$ - If φ is of the form $\neg\varphi, \forall x : \psi, \exists x : \psi$, then $subformulas(\varphi) = \{\varphi\} \cup subformulas(\psi)$ - If φ is of the from $\psi_1 \odot \psi_2$ where $\odot \in \{\vee, \wedge, \rightarrow\}$ then $subformulas(\varphi) = \{\varphi\} \cup subformulas(\psi_1) \cup subformulas(\psi_2)$			
A formula $\psi \in subformulas(\varphi)$ is called a proper subformula of φ , if $\varphi \neq \psi$.			
Binding Strength (to avoid parentheses)			
1. $\neg, \forall, \exists, \neg, \neg, 2. \wedge, 3. \vee, 4. \rightarrow$			
Example : $\neg A \wedge \exists x B \rightarrow C \vee D \leftrightarrow E$ is read as: $((\neg A) \wedge (\exists x B)) \rightarrow (C \vee D) \leftrightarrow E$			
Quantifier scope and bound variables			
Given a formula $\mathcal{Q}x : \psi$ where $\mathcal{Q} \in \{\exists, \forall\}$: - variable $x \in \mathcal{V}$ is bound in ψ ψ is the scope of x			
The quantifier only binds the first occurrence of a variable if outside a parentheses.			
Example: In the formula $\exists x P(x) \wedge R(x)$ only the first occurrence of x is bound, the second occurrence is free			
Given a formula tree of a formula, a variable x is bound if it has an upward path to $\mathcal{Q}x$.			
$\forall x((P(x) \rightarrow \neg Q(x)) \wedge (S(x, f(y, z)) \vee T(y)))$			

Entailment : (Question from exam - own solution)

Does the following entailment hold?

$$\{\exists x : F(x), \exists x, y : (R(x, y) \wedge G(y)), \forall x, y : ((F(x) \wedge R(x, y)) \rightarrow \neg G(y))\} \models \exists x, y : (F(x) \wedge R(x, y))$$

Give a first-order tableau proof or provide a countermodel.

$A \models B$ holds if $A \wedge \neg B$ is unsat (proved using tableau)

t : $\exists x : F(x)$
t : $\exists x, y : (R(x, y) \wedge G(y))$
t : $\forall x, y : ((F(x) \wedge R(x, y)) \rightarrow \neg G(y))$
f : $\exists x, y : (F(x) \wedge R(x, y))$

t : $F(c_1)$
t : $R(c_2, c_3) \wedge G(r_3)$
t : $R(c_2, c_3)$
t : $G(r_3)$

t : $F(c_1) \wedge R(c_1, c_r) \rightarrow \neg G(r_3)$

f : $F(c_1) \wedge R(c_1, c_3)$
f : $F(c_1)$
f : $R(c_1, c_3)$

f : $F(c_1) \wedge R(c_1, c_3)$
f : ...
//remaining formulas derived from
f : $\exists x, y : F(x) \wedge R(x, y)$

f : $F(c_1) \wedge R(c_1, c_3)$
f : $F(c_1)$
f : $R(c_1, c_3)$
//no more inference rules can be
applied to this branch and it didn't close
==> negation of entailment IS satisfiable

Using the open branch, one can create the following counterexample to the entailment:

$\mathcal{D} = \{c_1, c_2, c_3\}$ //Recall: the syntax of an
 $I(F) = \{\{c\}\}$ interpretation is on the other page
 $I(G) = \{\{c_3\}\}$
 $I(R) = \{\{c_2, c_3\}\}$

Formalization (1) : (Question from exercise sheet - sample solution)

Formulate the following sentences:

1. All dogs howl at night.
2. Someone who has cat will not have any mice.
3. Light sleepers do not have anything which howls at night.
4. John has either a cat or a dog.

Using the constant *John* to represent John and the predicates

$\neg Dog(x) := x$ is a dog,
 $\neg Howl(x) := x$ howls at night.
 $\neg Have(x, y) := x$ has y .
 $\neg Cat(x) := x$ is a cat.
 $\neg Mice(x) := x$ is a mice.
 $\neg LS(x) := x$ is a light sleeper.

we can formulate the given sentences as:

1. $\forall x : (Dog(x) \rightarrow Howl(x))$
2. $\forall x, y : (Have(x, y) \wedge Cat(y) \rightarrow \neg \exists z : (Have(x, z) \wedge Mice(z)))$
3. $\forall x : (LS(x) \rightarrow \neg \exists y : (Have(x, y) \wedge Howl(y)))$
4. $\exists x : (Have(John, x) \wedge (Cat(x) \vee Dog(x)))$

Formalization (2) : (Question from exercise sheet - sample solution)

Formulate the following sentences in first-order logic:

1. All actors and journalists invited to the party are late.
2. There is at least one person on time.
3. There is at least one invited person who is neither a journalist nor an actor.

We use the predicates

$\neg Actor(x) := x$ is an actor.
 $\neg Journalist(x) := x$ is a journalist.
 $\neg Invited(x) := x$ is invited
 $\neg Late(x) := x$ is late.

The above sentences can be formalised as follows

1. $\forall x : ((Actor(x) \vee Journalist(x)) \wedge Invited(x) \rightarrow Late(x))$
2. $\exists x : (\neg Late(x))$
3. $\exists x : (\neg(Actor(x) \vee Journalist(x)) \wedge Invited(x))$

Prenex Normal Form Transformation : (Question from exercise sheet - sample solution)

Convert the following FOL formulas into pnf:

$$\begin{aligned}\psi_1 : & \quad \forall x(P(x) \rightarrow \exists yQ(x, y)) \wedge \exists xP(x) \\ \psi_2 : & \quad \exists x(P(x) \wedge \forall y(R(y) \rightarrow Q(x, y)) \wedge \neg(\exists yQ(y, x))) \\ \psi_3 : & \quad \neg(\forall x\exists y(\neg P(y) \vee Q(x, f(y)) \vee Q(y, z)))\end{aligned}$$

• The case of ψ_1 :

1. Eliminate $\rightarrow$: $\forall x(\neg P(x) \vee \exists yQ(x, y)) \wedge \exists xP(x)$;
2. Move all negations inwards to the atomic level (Nothing need to be done):
 $\forall x(\neg P(x) \vee \exists yQ(x, y)) \wedge \exists xP(x)$;
3. Rename all variables apart: $\forall x_1(\neg P(x_1) \vee \exists x_2Q(x_1, x_2)) \wedge \exists x_3P(x_3)$;
4. Move all quantifiers to the front:
(a) $\forall x_1\exists x_2(\neg P(x_1) \vee Q(x_1, x_2)) \wedge \exists x_3P(x_3)$;
(b) $\forall x_1(\exists x_2(\neg P(x_1) \vee Q(x_1, x_2)) \wedge \exists x_3P(x_3))$;
(c) $\forall x_1\exists x_2((\neg P(x_1) \vee Q(x_1, x_2)) \wedge \exists x_3P(x_3))$;
(d) $\forall x_1\exists x_2\exists x_3((\neg P(x_1) \vee Q(x_1, x_2)) \wedge P(x_3))$.

• The case of ψ_2 :

1. Eliminate $\rightarrow$: $\exists x(P(x) \wedge \forall y(\neg R(y) \vee Q(x, y)) \wedge \neg(\exists yQ(y, x)))$;
2. Move all negations inwards to the atomic level: $\exists x(P(x) \wedge \forall y(\neg R(y) \vee Q(x, y)) \wedge \forall y\neg Q(y, x))$;
3. Rename all variables apart: $\exists x_1(P(x_1) \wedge \forall x_2(\neg R(x_2) \vee Q(x_1, x_2)) \wedge \forall x_3\neg Q(x_3, x_1))$;
4. Move all quantifiers to the front:
(a) $\exists x_1\forall x_2(P(x_1) \wedge (\neg R(x_2) \vee Q(x_1, x_2)) \wedge \forall x_3\neg Q(x_3, x_1))$;
(b) $\exists x_1\forall x_2\forall x_3(P(x_1) \wedge (\neg R(x_2) \vee Q(x_1, x_2)) \wedge \neg Q(x_3, x_1))$.

• The case of ψ_3 :

1. Eliminate $\rightarrow$ (Nothing need to be done): $\neg(\forall x\exists y(\neg P(y) \vee Q(x, f(y)) \vee Q(y, z)))$;
2. Move all negations inwards to the atomic level:
(a) $\exists x\forall y\neg(\neg P(y) \vee Q(x, f(y)) \vee Q(y, z))$;
(b) $\exists x\forall y(P(y) \wedge \neg Q(x, f(y)) \wedge \neg Q(y, z))$;
3. Rename all variables apart: $\exists x_1\forall x_2(P(x_2) \wedge \neg Q(x_1, f(x_2)) \wedge \neg Q(x_2, x_3))$;
4. Move all quantifiers to the front (Nothing need to be done):
 $\exists x_1\forall x_2(P(x_2) \wedge \neg Q(x_1, f(x_2)) \wedge \neg Q(x_2, x_3))$.

Theory consistency : (Question from exercise sheet - sample solution)

Consider the following formulas.

$$\begin{aligned}\varphi_1 &= \forall x\exists yR(x, y) & \varphi_2 &= \forall x\forall y(R(x, y) \rightarrow R(y, x)) \\ \psi_1 &= \exists x\forall yR(x, y) & \psi_2 &= \forall x\forall y\forall z((R(x, y) \wedge R(y, z)) \rightarrow R(x, z)).\end{aligned}$$

Answer the questions.

(a) Is the theory $\{\varphi_1, \varphi_2, \psi_1\}$ consistent? (or equivalently, is $\varphi_1 \wedge \varphi_2 \wedge \psi_1$ satisfiable?)

(b) What about the theory $\{\neg\varphi_1, \varphi_2, \psi_1\}$, is it consistent?

(c) Finally, consider the theory $\{\varphi_1, \varphi_2, \neg\psi_1, \neg\psi_2\}$. Is it consistent?

(a) Let $\mathcal{M} = (D, I)$ be a structure and $\ell : V \rightarrow D$ a variable assignment for $\mathcal{M}$ that

- $D = \{0, 1\}$ and $R^{\mathcal{M}} = I(R) = \{(0, 0), (0, 1), (1, 0)\}$.

So $\{\varphi_1, \varphi_2, \psi_1\}$ is satisfiable in $\mathcal{M}$ wrt ℓ :

- $R^{\mathcal{M}}$ is serial and symmetric, so φ_1 and φ_2 are satisfiable in $\mathcal{M}$ wrt ℓ ;
- ψ_1 is satisfiable — there is a $0 \in D$ such that $(0, 1)$ and $(0, 0)$ are both in $R^{\mathcal{M}}$.

(b) Suppose $\{\neg\varphi_1, \varphi_2, \psi_1\}$ is satisfiable in $\mathcal{M} = (D, I)$ wrt ℓ that $R^{\mathcal{M}} = I(R)$.

- From $\neg\varphi_1$ is satisfiable, there is a $d \in D$ for all $c \in D$ that $(d, c) \notin R^{\mathcal{M}}$ ($@_1$).
- From ψ_1 is satisfiable, there is a $d' \in D$ for all $c \in D$ that $(d', c) \in R^{\mathcal{M}}$ ($@_2$).
- From φ_2 is satisfiable in $\mathcal{M}$ wrt ℓ , $R^{\mathcal{M}}$ is symmetric. In addition with ($@_2$), which gives $(d', d) \in R^{\mathcal{M}}$, so $(d, d') \in R^{\mathcal{M}}$. However, this result is in conflict to ($@_1$).

So $\{\neg\varphi_1, \varphi_2, \psi_1\}$ is **not** satisfiable.

(c) Let $\mathcal{M} = (D, I)$ be a structure and $\ell : V \rightarrow D$ a variable assignment for $\mathcal{M}$ that

- $D = \{0, 1\}$ and $R^{\mathcal{M}} = I(R) = \{(0, 1), (1, 0)\}$.

So $\{\varphi_1, \varphi_2, \neg\psi_1, \neg\psi_2\}$ is satisfiable in $\mathcal{M}$ wrt ℓ :

- As $R^{\mathcal{M}}$ is serial and symmetric, φ_1 and φ_2 are satisfiable in $\mathcal{M}$ wrt ℓ ;
- From $(0, 0), (1, 1) \notin R^{\mathcal{M}}$, we know $\neg\psi_1$ is satisfiable in $\mathcal{M}$ wrt ℓ ;
- From $R^{\mathcal{M}}$ is not transitive — $(0, 1), (1, 0) \in R^{\mathcal{M}}$ but $(0, 0) \notin R^{\mathcal{M}}$, it follows that $\neg\psi_2$ is satisfiable in $\mathcal{M}$ wrt ℓ .

Finding a model : (Question from exercise sheet - sample solution)

Consider the following formulas.

$$\begin{aligned}\varphi_1 &= \forall x : R(x, s(x)) \\ \varphi_2 &= \forall x, y : (R(x, y) \rightarrow \neg R(y, x)) \\ \varphi_3 &= \forall x, y, z : (R(x, y) \wedge R(y, z) \rightarrow R(x, z))\end{aligned}$$

- a) Can you find a model of $\varphi_1 \wedge \varphi_2 \wedge \varphi_3$?
- b) Is there a model of $\varphi_1 \wedge \varphi_2 \wedge \varphi_3$ with a finite domain? Explain your answer!

(a) Let $\mathcal{M} = (D, I)$ be a structure and $\ell : V \rightarrow D$ a variable assignment for $\mathcal{M}$ that

- $D = \mathbb{N}$ and $I(R) = <$, as well as $s = I(s) : \mathbb{N} \rightarrow \mathbb{N}$ such that $s(n) = n + 1$ where $n \in \mathbb{N}$.

So $\varphi_1 \wedge \varphi_2 \wedge \varphi_3$ is satisfiable in $\mathcal{M}$:

- φ_1 is satisfiable in $\mathcal{M}$ wrt ℓ — $(n, n + 1) \in <$ for all $n \in \mathbb{N}$.
- As $<$ is asymmetric and transitive, therefore the model $\mathcal{M}$ satisfies φ_2 and φ_3 as well.

(b) Suppose there is a finite model $\mathcal{M} = (D, I)$ and $\ell : V \rightarrow D$ a variable assignment for $\mathcal{M}$ such that $D = \{d_1, \dots, d_n\}$, $R^{\mathcal{M}} = R \subseteq D \times D$, and $s^{\mathcal{M}} = s : D \rightarrow D$. Assume that $\{\varphi_1, \varphi_2, \varphi_3\}$ is satisfiable in $\mathcal{M}$ wrt ℓ . So,

- From $\mathcal{M}, \ell \models \varphi_1$ it follows that for all $d \in D$ that $dRs(d)$ ($@_1$);
- From $\mathcal{M}, \ell \models \varphi_2$ it follows that for all $d, c \in D$ that if dRc then **not** cRd ($@_2$).
- From $\mathcal{M}, \ell \models \varphi_3$ it follows that for all $d, c, e \in D$ that if dRc and cRe then dRe ($@_3$).

- We define a new notation $s^{n+1}(d) = \overbrace{s(\dots s(d))}^{n+1} = s^n(d)$ — so, $s^n(d)$ means that we apply the function s on $d \in D$ in n times.
- By ($@_1$), it follow that $s^n(d)Rs^{n+1}(d)$ where $n \in \mathbb{N}$ ($@_4$) — meaning that every $s^n(d)$ is accessed by its direct successor $s^{n+1}(d)$.
- By ($@_3$) and ($@_4$), we have $dRs^k(d)$ ($@_5$) — meaning that every d is accessed by its k -successor $s^k(d)$, where $k \in \mathbb{N}$.
- By ($@_2$), given any dRc , it is not possible to have $d = c$. Otherwise, it would lead to dRd which is in conflict to ($@_2$). This means that, every point cannot be its successor nor its ancestor ($@_6$).
- By ($@_5$) and ($@_6$), we have $dRs^1(d), dRs^2(d), \dots, dRs^n(d)$; however, $d, s^1(d), s^2(d), \dots, s^n(d)$ are $n + 1$ different points, which is contrary to the assumption that D only have n elements!

- From ($@_4$), we can enumerate $dRs^1(d), s^1(d)Rs^2(d), \dots, s^{k-1}(d)Rs^k(d)$, where $k < n$.
- Otherwise, $k \geq n$. Suppose $k = n$ and so $s^{n-1}(d)Rs^n(d)$.
- Given that D only have n elements, there must be a $i < n$ such that $s^n(d) = s^i(d)$. So $s^{n-1}(d)Rs^i(d)$ ($@_7$).
- By ($@_3$) and the enumeration, we have $s^i(d)Rs^{n-1}(d)$ ($@_8$).
- However, ($@_7$) and ($@_8$) together are in conflict to ($@_2$).